

CSCE-636: Homework 2

Due on September 24, 2025 at 11:59 pm

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Problem 1

Check out the Jupyter notebook for Chapter 3 at https://github.com/fchollet/deep-learning-with-python-notebooks/blob/master/chapter03_introduction-to-ml-frameworks.ipynb. Then, use the "GradientTape API" to find the derivative of the function $f(x) = \sin(x)$ for $x = 0, 0.1, 0.2$ and 0.3 . Submit your Jupyter notebook that shows both the code and the result you got.

Solution

Problem 2

Check out the Jupyter notebook for Chapter 4 at https://github.com/fchollet/deep-learning-with-python-notebooks/blob/master/chapter04_classification-and-regression.ipynb. Then, for the task "Classifying movie reviews: A binary classification example", tune the hyper-parameters of the model (such as changing the number of layers, changing the sizes of layers, changing the optimizer, changing the learning rate, etc.), and see if you can improve the model's performance. Submit a Jupyter notebook where you clearly show the code with the best hyper-parameters that you have found, along with its performance on training, validation and test sets.

Solution